

This Booklet contains 20 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marketing responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

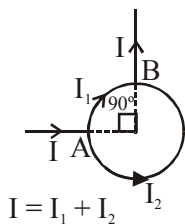
Facsimile signature stamp of

Centre Superintendent : _____

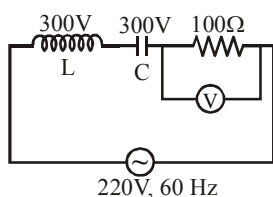
LEADER & ACHIEVER COURSE
**(PHASE : MLI,J,K,M,N,O,R,S,M4L8A,MAZG,H,I,J,K,L,M,T,U,M1A5A,
M2A5B,M4AA2A,M2AP1A,M2AP1B)**

ALLEN MAJOR TEST		DATE : 23 - 03 - 2020
SYLLABUS – 05 & 06		
PHYSICS	:	Magnetic effect of current and Magnetism, Electromagnetic Induction, Alternating current, EM Waves, Ray optics and optical Instruments, Wave optics, Modern Physics, Semiconductor and Digital Electronics.
CHEMISTRY	:	p-block elements (group 13 to 18), d & f-block elements Coordination Compounds, Metallurgy, Halogen derivatives, Oxygen containing organic compounds, Nitrogen containing organic compounds, Biomolecules, polymers and chemistry in everyday life
BIOLOGY	:	Genetics and Evolution : (i) Principles of inheritance and Variation (ii) Molecular basis of inheritance Biology in Human Welfare : (i) Microbes in Human Welfare Biotechnology : (i) Biotechnology : Principles and Processes (ii) Biotechnology and its Applications (iii) DOP Biology in Human Welfare : (i) Human Health and Disease (ii) DOA Ecology : (i) Organisms and Populations (ii) Ecosystem (iii) Biodiversity and Conservation (iv) Environmental Issues

1. A current I enters a circular coil of radius R , branches into two parts and then recombines as shown in the circuit diagram. The resultant magnetic field at the centre of the coil is :-

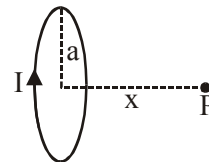


- (1) zero (2) $\frac{\mu_0 I}{2R}$
- (3) $\frac{3}{4} \left(\frac{\mu_0 I}{2R} \right)$ (4) $\frac{1}{4} \left(\frac{\mu_0 I}{2R} \right)$
2. A magnetic field of 2×10^{-2} tesla acts at right angles to a coil of area 100 cm^2 with 50 turns. The average e.m.f. induced in the coil is 0.1 V when it is removed from the field in time t . The value of t is:
- (1) 0.1 sec (2) 0.01 sec
- (3) 1 sec (4) 20 sec
3. In the circuit shown below, reading of the voltmeter will be :-



- (1) 200 V (2) 800 V
- (3) 100 V (4) 220 V
4. In a transistor in CE configuration, the ratio of power gain to voltage gain is :-
- (1) α (2) β/α
- (3) $\beta \times \beta$ (4) β

5. By a radioactive material :-
- (1) α , β and γ -rays are emitted together
- (2) α and β are emitted together and there after γ is emitted
- (3) α , β and γ are emitted one by one
- (4) γ -rays are emitted after the emission of α or β particles
6. The magnetic field at a point P due to a current carrying coil is given by :-

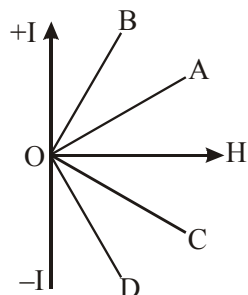


- (1) $\frac{1}{4\pi\epsilon_0} \frac{I}{a^2}$ (2) $\frac{\mu_0}{4\pi} \frac{Iax}{(x^2 - a^2)^2}$
- (3) $\frac{1}{4\pi\epsilon_0} \cdot \frac{2\pi Ia}{x^2}$ (4) $\frac{\mu_0}{4\pi} \frac{2\pi Ia^2}{(x^2 + a^2)^{3/2}}$
7. The magnetic flux linked with a circuit of resistance 100 ohm increases from 10 to 60 webers. The amount of induced charge that flows in the circuit is: (in coulomb) :-
- (1) 0.5 (2) 5
- (3) 50 (4) 100
8. If $V = 100 \sin 100t$ volt, and
- $I = 100 \sin(100t + \frac{\pi}{6})$ A. then find the watt less power in watt :-
- (1) 10^4 (2) 10^3
- (3) 10^2 (4) 2.5×10^3
9. In an n-p-n transistor, 10^8 electrons enter the emitter in 10^{-8} s. If 1% electrons are lost in the base, the fraction of current that enters the collector and current amplification factor (β) are respectively:-
- (1) 0.8 and 49
- (2) 0.9 and 90
- (3) 0.7 and 50
- (4) 0.99 and 99

10. A radio transmitter operates at a frequency of 880 kHz and a power of 10 kW. The number of photons emitted per second are :-

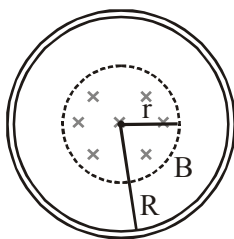
- (1) 1.72×10^{31}
- (2) 1327×10^{34}
- (3) 13.27×10^{34}
- (4) 0.075×10^{-34}

11. The variation of the intensity of magnetisation (I) with respect to the magnetising field (H) in a diamagnetic substance is described by the graph :-



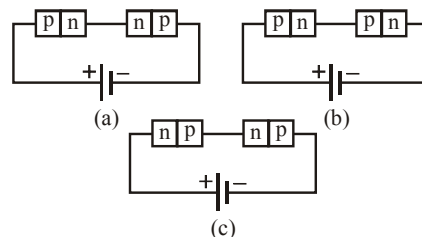
- (1) OD
- (2) OC
- (3) OB
- (4) OA

12. Figure shows a conducting ring of radius R. A uniform steady magnetic field B lies perpendicular to the plane of the ring in a circular region of radius r ($r < R$). If the resistance per unit length of the ring is λ , then the current induced in the ring when its radius gets doubled is :



- (1) $\frac{BR}{\lambda}$
- (2) $\frac{2BR}{\lambda}$
- (3) Zero
- (4) $\frac{BR^2}{4R\lambda}$

13. The p-n junctions can be connected in series by three methods as shown in the figure given below. If the potential difference in the junctions is the same, then the correct connections will be :-



- (1) in circuits (a) and (b)
- (2) in circuits (b) and (c)
- (3) in circuits (a) and (c)
- (4) only in the circuit (a)

14. NOR gate is a combination of :-

- (1) OR gate and NOT gate
- (2) OR gate and AND gate
- (3) OR gate and OR gate
- (4) None of the above

15. A person's eye level is at 1.5 m. He stands in front of a 0.3 m long plane mirror which is kept on ground. The length of image he sees of himself is :

- (1) 0.6 m
- (2) 0.15 m
- (3) 0.75 m
- (4) zero

16. In a mass spectrograph O^{++} , He^{++} and H^+ are projected on a photographic plate with same momentum in uniform magnetic field then which will strike the plate farthest :-

- (1) O^{++}
- (2) He^{++}
- (3) Both (1) and (2)
- (4) H^+

17. Induced electric field due to time varying magnetic field is -

- (1) Conservative
- (2) Non Conservative
- (3) Both Conservative and Non Conservative
- (4) None of these

18. In a semiconducting material $\left(\frac{1}{5}\right)$ th of the total current is carried by the holes and the remaining is carried by the electrons. The drift speed of electrons is twice that of holes at this temperature, the ratio between the number densities of electrons and holes is :-

(1) $\frac{21}{6}$ (2) 5 (3) $\frac{3}{8}$ (4) 2

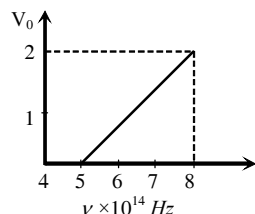
19. The stopping potential (V_0) versus frequency (ν) plot of a substance is shown in figure the threshold wavelength is :-

(1) 5×10^{14} m

(2) $6000\text{\AA}$

(3) $5000\text{\AA}$

(4) Cannot be estimated from given data

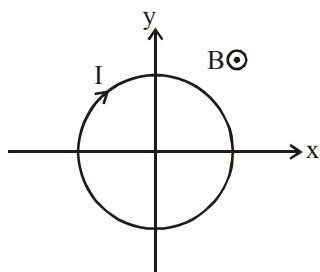


20. A concave mirror of focal length of (in air) is f ,

immersed in water $\left(\mu = \frac{4}{3}\right)$. The focal length of mirror in water will be :-

(1) f (2) $\frac{4}{3}f$ (3) $\frac{3}{4}f$ (4) $\frac{7}{3}f$

21. A conducting loop carrying a current I is placed in a uniform magnetic field as shown. The loop will have a tendency to :-



- (1) Contract
(2) Expand
(3) Move toward +ve x-axis
(4) Move toward (-ve) x-axis

22. A step down transformer is used to reduce the main supply of 220V to 11V. If the primary coil draws a current of 5A and the current in secondary coil 90A, what is the efficiency of the transformer?

(1) 60 %
(2) 70%
(3) 80%
(4) 90%

23. The maximum efficiency of full wave rectifier is :-

(1) 100%
(2) 25.20%
(3) 40.2%
(4) 81.2%

24. There are n_1 photons of frequency γ_1 in a beam of light. In an equally energetic beam, there are n_2 photons of frequency γ_2 . Then the correct relation is :-

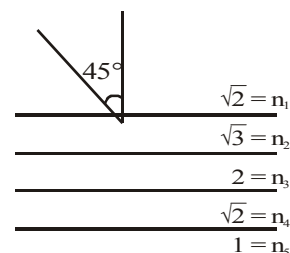
(1) $\frac{n_1}{n_2} = 1$

(2) $\frac{n_1}{n_2} = \frac{\gamma_1}{\gamma_2}$

(3) $\frac{n_1}{n_2} = \frac{\gamma_2}{\gamma_1}$

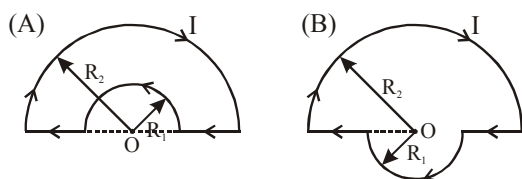
(4) $\frac{n_1}{n_2} = \frac{\gamma_1^2}{\gamma_2^2}$

25. In the figure shown the angle made by the light ray with the normal in the medium having index n_4 is :



- (1) 30°
(2) 60°
(3) 45°
(4) None of these

26. The wire loop PQRSP formed by joining two semicircular wires of radii R_1 and R_2 carries a current I as shown in fig. What is the ratio of magnetic moment of the loop in cases (A) and (B) respectively :-

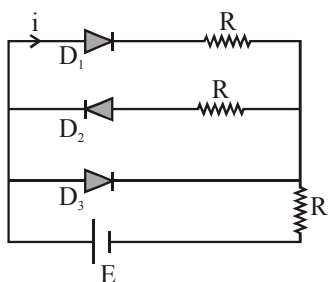


- (1) $\frac{R_2 - R_1}{R_1 + R_1}$ (2) $\frac{R_1^2}{R_2^2}$
 (3) $\frac{R_2 + R_1}{R_2 - R_1}$ (4) $\frac{R_2^2 - R_1^2}{R_2^2 + R_1^2}$

27. An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$; $B = B_0 \sin(kx - \omega t)$. Which of the following equations is true ?

- (1) $E_0 k = B_0 \omega$ (2) $E_0 \omega = B_0 k$
 (3) $E_0 B_0 = \omega k$ (4) None of these

28. In the following circuit of PN junction diodes D_1 , D_2 and D_3 are ideal then i is :-



- (1) E/R (2) $E/2R$
 (3) $2E/3R$ (4) Zero

29. The de Broglie wavelength λ associated with a proton changes by 0.25% if its momentum is changed by p_0 . The initial momentum was :-

- (1) $100 p_0$ (2) $p_0/400$
 (3) $400 p_0$ (4) $p_0/100$

30. In an electron microscope we use :

- (1) Convex lens of flint glass
 (2) Concave lens of crown glass
 (3) Magnetic lens
 (4) We use borax glass lens

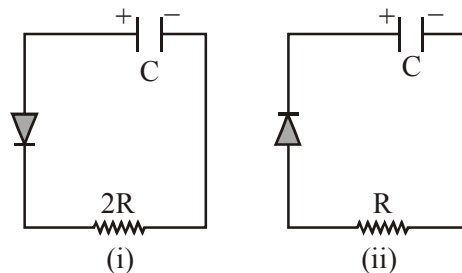
31. A magnet of magnetic moment M is situated with its axis along the direction of a magnetic field strength B . The work done in rotating it by an angle of 180° will be :

- (1) $-MB$ (2) $+MB$
 (3) 0 (4) $+2MB$

32. Find the values of magnetic field between plates of capacitor at distance 1 m from centre, where electric field varies by 10^{10} V/m per second :-

- (1) 5.56×10^{-8} T (2) 5.56×10^{-3} T
 (3) $5.56 \mu\text{T}$ (4) 5.56 T

33. Two identical capacitors A and B are charged to the same potential V and are connected in two circuit at $t = 0$ as shown in figure. The charge on the capacitor at time $t = RC$ are respectively :-

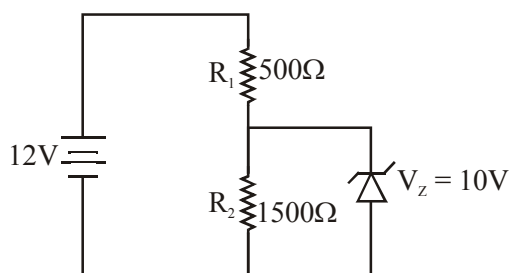


- (1) $\frac{CV}{2e}, CV$ (2) $\frac{CV}{e}, CV$
 (3) $\frac{CV}{e^2}, 0$ (4) $\frac{CV}{e^{1/2}}, CV$

34. The rate of disintegration was observed to be 10^{17} disintegrations per sec. Its half life period is 1445 years. The number of particles are:-

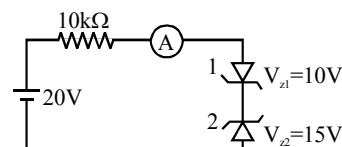
- (1) 8.9×10^{27}
 (2) 6.6×10^{27}
 (3) 1.4×10^{16}
 (4) 1.2×10^{17}

35. In the far field diffraction pattern of a single slit under polychromatic illumination, the first minimum with the wavelength λ_1 is found to be coincide with the third maximum λ_2 . So
 (1) $3\lambda_1 = 0.3\lambda_2$
 (2) $3\lambda_1 = \lambda_2$
 (3) $\lambda_1 = 3.5\lambda_2$
 (4) $0.3\lambda_1 = 3\lambda_2$
36. The magnetic potential at a point on the axial line of magnet of dipole moment M is V . What is the magnetic potential due to a bar magnet of dipole moment $\frac{M}{4}$ at the same point ?
 (1) $4V$ (2) $2V$ (3) $V/2$ (4) $V/4$
37. A resistance of $20\ \Omega$ is connected to a source of an alternating potential $V = 220 \sin(100\ \pi t)$. The time taken by the current to change from the peak value to rms value, is
 (1) $0.2\ s$
 (2) $0.25\ s$
 (3) $2.5 \times 10^{-4}\ s$
 (4) $2.5 \times 10^{-3}\ s$
38. In the given circuit the current through the Zener diode is close to :-



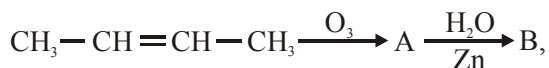
- (1) $6\ mA$ (2) $4\ mA$
 (3) $6.7\ mA$ (4) $0\ mA$
39. A heavy nucleus at rest breaks into two fragments which fly off with velocities in the ratio $8 : 1$. The ratio of radii of the fragments is :-
 (1) $1 : 2$ (2) $1 : 4$
 (3) $4 : 1$ (4) $2 : 1$

40. Three polaroids are kept coaxially. Angle between the first and third polaroid is 90° . Angle between the first and second polaroid is 60° . If unpolarized light energy incident on the first polaroid is I_0 . Light energy that emerges from the system is
 (1) zero (2) $\frac{3I_0}{32}$
 (3) $\frac{3I_0}{16}$ (4) $\frac{\sqrt{3}I_0}{8}$
41. At a certain place a magnet makes 30 oscillations per minute. At another place where the magnetic field is double its time period will be :
 (1) $4\ sec$ (2) $2\ sec$
 (3) $1/2\ sec$ (4) $\sqrt{2}\ sec$
42. A coil of inductance $0.015\ henry$ and resistance $1\ \Omega$ is connected in series with $3\ resistance$. If the current is $2\ Amp.$ and e.m.f. is $10\ volt$ in the circuit then reactance of the coil will be :
 (1) $2\ \Omega$ (2) $3\ \Omega$
 (3) $4\ \Omega$ (4) $5\ \Omega$
43. The reading of ammeter in the following circuit.



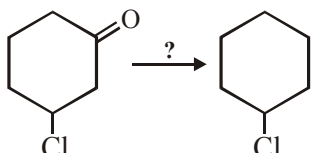
- (1) $0.5\ mA$ (2) $1.0\ mA$
 (3) $1.5\ mA$ (4) $2.5\ mA$
44. The half life of a radioactive material is T . If the activity of that material in the beginning was A_0 and after a time t the activity was A , then:
 (1) $A = A_0 2^{t/T}$ (2) $A = A_0 2^{tT}$
 (3) $A_0 = A 2^{t/T}$ (4) $A = A_0 2^{-t/T}$
45. In Young's experiment with sodium light, the slits are $0.589\ m$ apart. What is the angular position of the third maximum ? (Given that $\lambda = 589\ nm$) :-
 (1) $\sin^{-1}(3 \times 10^{-6})$
 (2) $\sin^{-1}(3 \times 10^{-8})$
 (3) $\sin^{-1}(0.33 \times 10^{-6})$
 (4) $\sin^{-1}(0.33 \times 10^{-8})$

46. In the following sequence of reaction; the alkene affords the compound 'B' :-



the compound B is :-

- (1) $\text{CH}_3-\text{CO}-\text{CH}_3$
 - (2) $\text{CH}_3-\text{CH}_2-\text{CO}-\text{CH}_3$
 - (3) CH_3-CHO
 - (4) $\text{CH}_3-\text{CH}_2-\text{CHO}$
47. Which of the reducing reagent will be most suitable for following conversion ?



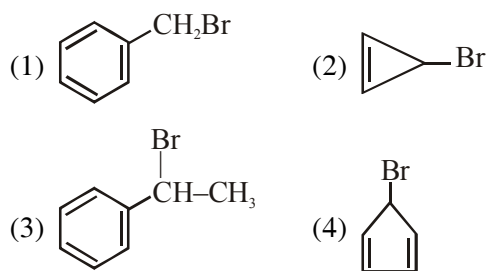
- (1) Zn-Hg/HCl
 - (2) $\text{NH}_2-\text{NH}_2/\text{OH}^\ominus$
 - (3) LiAlH_4
 - (4) NaBH_4
48. p-chloro aniline and anilinium hydrogen chloride can be distinguished by :-
- (1) Sandmeyer reaction
 - (2) Carbyl amine reaction
 - (3) Hinsberg's reaction
 - (4) AgNO_3
49. Which of the followings is not incorrect ?
- (1) CO is absorbed by ammonical Cu_2Cl_2
 - (2) Borax bead test not given by aluminium salt.
 - (3) no. of possible isomers for disubstituted borazine $\text{B}_3\text{N}_3\text{H}_4\text{X}_2$ is = 4
 - (4) All are correct
50. The color of light absorbed by an aqueous solutions of CuSO_4 is :-
- (1) orange red
 - (2) blue green
 - (3) yellow
 - (4) violet

51. The synthesis of ethene from electrolysis of an aqueous solution of potassium succinate is known as-
- (1) Faradays Electrolysis
 - (2) Kolbe - Schmidt reaction
 - (3) Hoffmann's rearrangement
 - (4) Kolbe's electrolytic synthesis

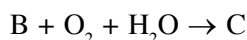
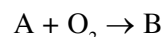
52. Anisol can be prepared by the action of methyl iodide on sodium phenoxide. The reaction is called.

- (1) Fitting reaction
- (2) Etard reaction
- (3) Williamson ether synthesis
- (4) Wurtz reaction

53. Which of the following will not give precipitate with aq. AgNO_3 :-

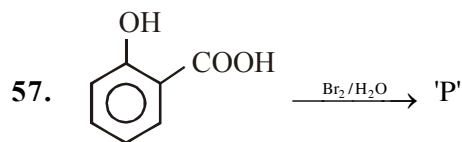
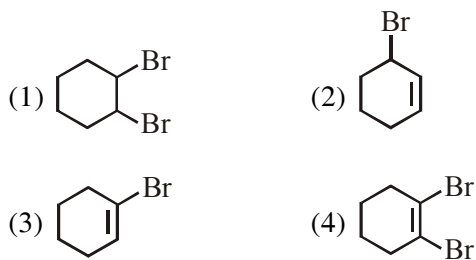
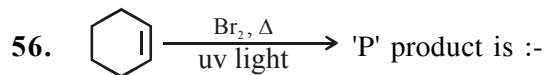


54. $\text{NH}_3 + \text{O}_2 \xrightarrow[\Delta]{\text{Pt}} \text{A} + \text{H}_2\text{O}$

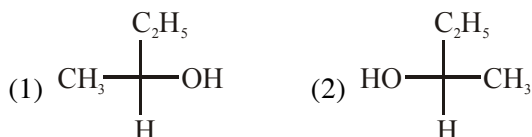
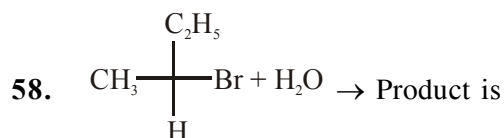
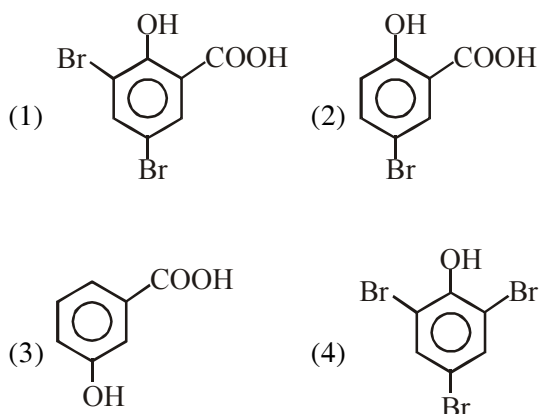


A, B & C are :-

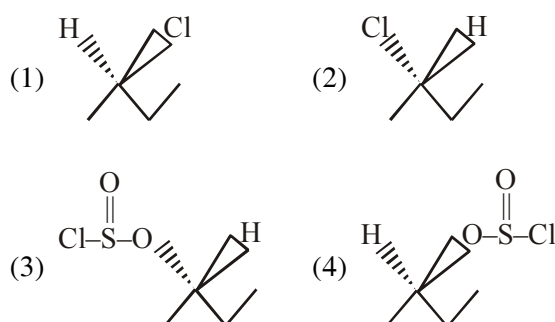
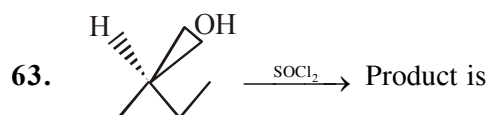
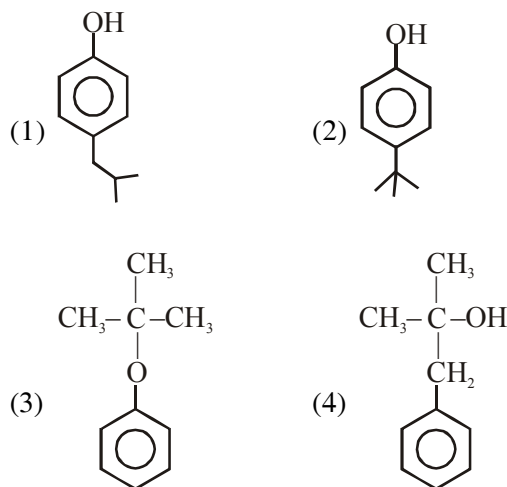
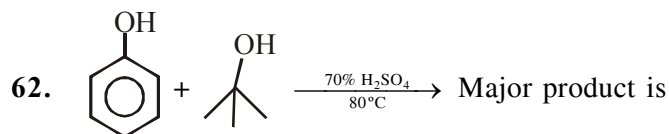
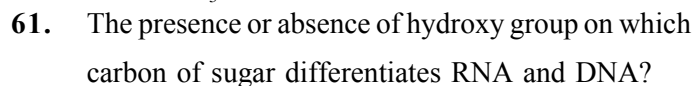
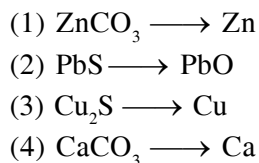
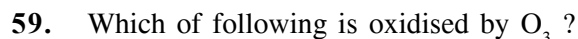
- (1) N_2O , NO_2 , HNO_3
 - (2) NO , NO_2 , HNO_3
 - (3) NO_2 , NO , HNO_3
 - (4) N_2O , NO , HNO_3
55. Which of the following has two geometrical isomers and is also a non ionisable complex :-
- (1) $\text{PtCl}_4 \cdot 2\text{NH}_3$
 - (2) $\text{PtCl}_4 \cdot 3\text{NH}_3$
 - (3) $\text{PtCl}_4 \cdot 4\text{NH}_3$
 - (4) $\text{PtCl}_4 \cdot 5\text{NH}_3$



Product P is :



(3) Mixture of 1 and 2 (4) None



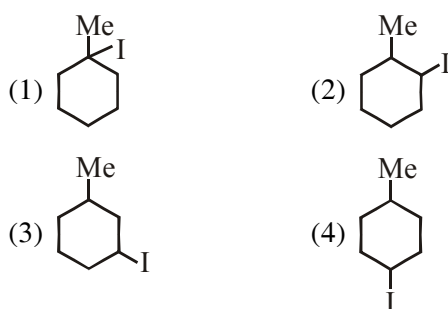
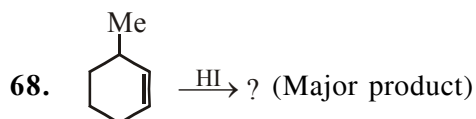
64. Most stable ion in aqueous solution is :-
 (1) Cr^{+3} (2) Mn^{+3} (3) Ti^{+3} (4) V^{+3}

65.

List-I		List-II	
(P)	$[\text{M}(\text{CN})_2]^-$	(A)	Ni (during refining Metal)
(Q)	$[\text{M}(\text{CO})_4]$	(B)	Ti (during refining Metal)
(R)	MI_4	(C)	Pb (during roasting of ore)
(S)	MO	(D)	Ag (during leaching of ore)

P	Q	R	S
(1) D	A	B	C
(2) D	B	A	C
(3) C	A	B	D
(4) A	B	D	C

66. Which of the following statement is incorrect :-
 (1) Acetals of Carbohydrate are called Glycoside
 (2) Sucrose does not form osazone
 (3) Anomeric carbon is also called Hemiacetal carbon
 (4) Starch is a polymer of β - D Glucose
67. Nitrobenzene combine with hydrogen in the presence of platinum to produce :-
 (1) Toluene (2) Benzene
 (3) Aniline (4) Azobenzene

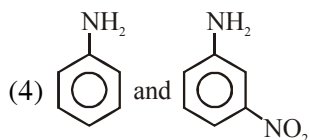
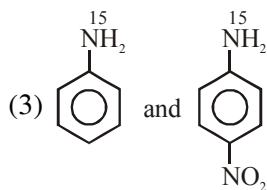
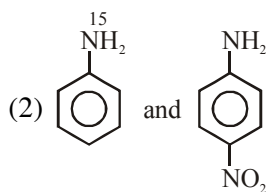
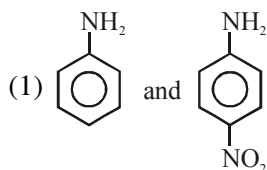
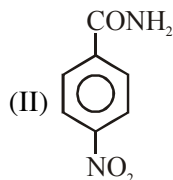
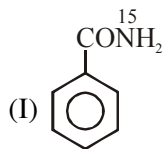


69. The C–O distances in $\text{V}(\text{CO})_6$ and $[\text{V}(\text{CO})_6]^-$ are respectively (in Pm) :-
 (1) 200, 200
 (2) 193, 200
 (3) 200, 193
 (4) 193, 193
70. Fac and Mer isomerism is associated with which of the following general formula ?
 (1) $[\text{M}(\text{AA})_2]$
 (2) $[\text{M}(\text{AA})_3]$
 (3) $[\text{MABCD}]$
 (4) $[\text{MA}_3\text{B}_3]$
71. Monomers of glyptal is -
 (1) Ethylene glycol and phthalic acid
 (2) Terephthalic acid and ethylene glycol
 (3) Hexamethylene tetramine and adipic acid
 (4) Phenol and formaldehyde
72. The well known explosive TNT stands for :-
 (1) 2,3,6-Trinitrotoluene
 (2) 3,4,5-Trinitrotoluene
 (3) 2,3,5-Trinitrotoluene
 (4) 2,4,6-Trinitrotoluene
73. Which of the following compound is diamagnetic but coloured in nature :-
 (1) K_2MnO_4
 (2) CrCl_3
 (3) $\text{Cr}_2\text{O}_7^{2-}$
 (4) MnO_2
74. $[\text{Pt}(\text{en})_3]_x [\text{PtCl}_4]_y$ possible value of $x + y$ in complex will be :-
 (1) 2 (2) 3 (3) 4 (4) 5
75. Which of the following will have two stereoisomeric forms ?
 (I) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (II) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 (III) $[\text{CoCl}_2(\text{en})_2]^+$ (IV) $[\text{CoBrCl}(\text{OX})_2]^{3-}$
 (1) I only
 (2) I and II
 (3) III and IV
 (4) All of these

76. Which of the following is semi-synthetic polymer ?

- (1) BUNA - S
- (2) PROTEINS
- (3) RAYON
- (4) NYLON - 6, 6

77. The products formed when a mixture of the following two amides (I and II) are treated with bromine and aqueous KOH are :-



78. Correct order of Reducing property is :-

- (a) $\text{SnCl}_2 > \text{PbCl}_2$
- (b) $\text{SnCl}_2 > \text{SnCl}_4$
- (c) $\text{SO}_2 > \text{SeO}_2$
- (d) $\text{TeO}_2 > \text{SeO}_2$
- (e) $\text{I}_2 > \text{F}_2$
- (1) a, b & e
- (2) a, b, c & d
- (3) a & b only
- (4) a, b, c & e

79. IUPAC name of $[\text{Co}(\text{NH}_3)_6][\text{Co}(\text{ONO})_6]$:-

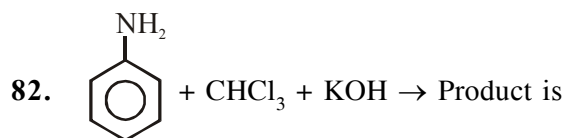
- (1) Hexaamminecobalt(II)
hexanitrito-Ncobaltate(II)
- (2) Hexaamminecobalt(III)
hexanitrito-Ncobaltate (III)
- (3) Hexaamminecobaltate(II)
hexanitrito-Ncobaltate(II)
- (4) Hexaamminecobalt(III)
hexanitrito-Ocobaltate(III)

80. For which of the following d^n configuration of octahedral complex, cannot exist in both high spin and low spin forms ?

- (1) d^3
- (2) d^5
- (3) d^6
- (4) d^4

81. Which is not Analgesics -

- (1) Paracetamol
- (2) Aspirin
- (3) Morphine
- (4) Tegamet



- (1) Phenyl Isocyanide
- (2) Benzyl amine
- (3) Benzyl chloride
- (4) Benzoic acid

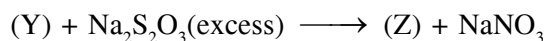
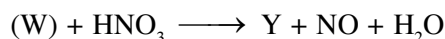
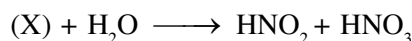
83. Which is pyrosilicate :-

- (1) $\text{Zn}_3\text{Si}_2\text{O}_7 \cdot \text{Zn}(\text{OH})_2$
- (2) ZrSiO_4
- (3) $\text{Mg}_3(\text{OH})_2(\text{Si}_2\text{O}_5)_2$
- (4) SiO_2

84. $[\text{NiCl}_4]^{-2}$ & $[\text{Ni}(\text{CN})_4]^{-2}$ shows similarity in :-

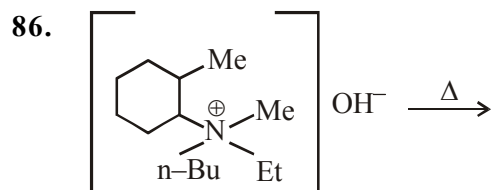
- (1) geometry
- (2) magnetic nature
- (3) hybridisation state of Ni
- (4) primary valency of Ni

85. $\text{AgNO}_3 \xrightarrow{\Delta} (\text{W}) + (\text{X}) + \text{O}_2$

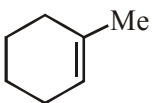
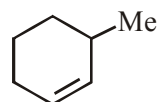
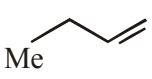


Identify (W) to (Z).

- | | |
|-------------------------------|--|
| (1) W = Ag | X = N_2O |
| Y = AgNO_3 | Z = $\text{Na}_2[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| (2) W = Ag_2O | X = NO |
| Y = AgNO_3 | Z = $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| (3) W = Ag | X = NO_2 |
| Y = AgNO_3 | Z = $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| (4) W = Ag_2O | X = N_2 |
| Y = AgNO_3 | Z = $\text{Na}[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |



The alkene formed as a major product in the above elimination reaction is :

- | | |
|---|---|
| (1) $\text{CH}_2 = \text{CH}_2$ | (2)  |
| (3)  | (4)  |

87. Which of the following undergoes mustard oil reaction :-

- (1) Primary amines
- (2) Secondary amines
- (3) Tertiary amines
- (4) All the above

88. For true statement use 'T' for false use 'F' ?

- (a) Extent of back bonding $\Rightarrow \text{O}(\text{SiH}_3)_2 > \text{PF}_3$
- (b) NH_3 is stronger lewis base than $\text{N}(\text{SiH}_3)_3$
- (c) CF_2 is more stable than CH_2 (Singlet)
- (d) H_3PO_4 can form three series of salts.

(1) T F T F

(2) T T T T

(3) F F F T

(4) F F T T

89. The CFSE value of octahedral high spin complex is zero. The spin magnetic moment can be :-

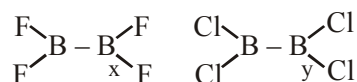
(1) 4.9 B.M.

(2) 3.87 B.M.

(3) 5.91 B.M.

(4) 1.72 B.M.

90. Compare the bond length in following structure-



(1) $x > y$

(2) $x = y$

(3) $x < y$

(4) Can not predict

91. How many diseases are there which have female mosquito as vector ?
Filariasis, Ringworm, Malaria, Typhoid, Pneumonia, Common Cold, Ascariasis, Amoebiasis
(1) Four (2) Two (3) Three (4) Five
92. Drugs which are commonly abused are obtained in majority from :-
(1) Algae (2) Fungi
(3) Flowering plants (4) Animals
93. Match the following and choose the correct option :-

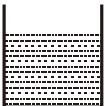
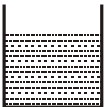
Column-I		Column-II	
a.	47, XXY	(i)	Klinefelter's syndrome
b.	Skin color in humans	(ii)	Polygenic inheritance
c.	ABO blood group in humans	(iii)	Incomplete dominance
d.	Starch grain size in seeds of garden pea	(iv)	Multiple alleles

- (1) a-i, b-ii, c-iii, d-iv (2) a-i, b-ii, c-iv, d-iii
(3) a-i, b-iii, c-ii, d-iv (4) a-ii, b-i, c-iii, d-iv
94. In mung bean, resistance to __A__ and __B__ were introduced by mutation breeding :-

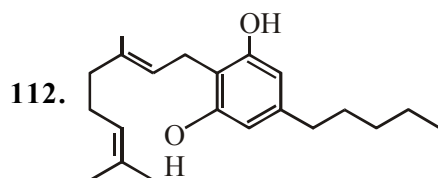
A	B
(1) Yellow mosaic virus	Powdery mildew
(2) Jassid	Aphids
(3) White rust	Hill bunt
(4) Bacterial blight	Leaf curl
95. In a stirred-tank bioreactor, the stirrer facilitates _____.
(1) Mixing of reactor contents
(2) Oxygen availability throughout the bioreactor
(3) Sterilisation
(4) Both (1) and (2)
96. Choose the wrong match :-
(1) Basmati rice – 27 documented varieties in India
(2) GEAC – Genetic Engineering Approval Committee
(3) Cotton bollworms-cry IAC
(4) α -Lactalbumin – 'Rosie' sheep

97. Biological control methods, which are used in malaria control, are based on which of the following property of predators :-
(1) Control of prey population
(2) Conduit of energy transfer across trophic levels
(3) Maintain species diversity in a community
(4) Prudent in nature
98. In an ecosystem, rate of formation of new organic matter by consumers is called as :-
(1) Gross Primary Productivity (GPP)
(2) Net Community Productivity (NCP)
(3) Net Primary Productivity (NPP)
(4) Secondary Productivity
99. Western Ghats have greater amphibian species diversity than the Eastern Ghats. This level of diversity is :-
(1) Genetic diversity (2) Species diversity
(3) Ecological diversity (4) All of the above
100. Which of the following is/are the harmful effect of photochemical smog ?
(1) Irritation in eyes
(2) Inhibition of photosystem-II
(3) Damage to mucous layer
(4) All of the above
101. Which of the following statement is correct ?
(1) Common cold infect lungs
(2) Hoarseness occurs in common cold
(3) Diphtheria is a viral disease
(4) Sporozoites enter human body through the bite of newly developed female anopheles
102. Cannabinoids includes :-
(1) Marijuana, Hashish only
(2) Marijuana, Hashish, Smack, Ganja
(3) Hashish, Marijuana, Charas, Ganja, Smack
(4) Hashish, Marijuana, Charas, Ganja
103. Which of the following N_2 -Bases is not found in mRNA ?
(1) Adenine (2) Uracil
(3) Guanine (4) Thymine
104. Which of the following bacteria is an example of single cell protein (SCP) ?
(1) Mushroom
(2) *Chlorella*
(3) *Methylophilus methylotrophus*
(4) Both (1) and (3)

105. Large holes in 'Swiss Cheese' are due to production of large amount of CO_2 by a bacterium named ____.
- Saccharomyces cerevisiae*
 - Propionibacterium sharmanii*
 - Clostridium butylicum*
 - Penicillium notatum*
106. The high quality soft wool "Shahtoosh" is obtained from:-
- Goat
 - Chiru
 - Rabbit
 - Sheep
107. Pacific salmon fish adopts which of the following strategy to maximise it's reproductive fitness :-
- Breeding many times in life
 - Breeding only once in life
 - Production of large number of small size offsprings
 - Both (2) and (3) given above
108. If a plant uses 3 percent of incident solar radiation (ISR) then percentage of Photosynthetically Active Radiation (PAR) that plant uses :-
- 3 Percent
 - 6 Percent
 - 1.5 Percent
 - 12 Percent
109. Conventional taxonomic methods are not suitable for identification of :-
- Insects
 - Fungi
 - Prokaryotes
 - Mammals
110. Which of the following sample of water is purest among the options given below ?

- (1)  BOD = 7
- (2)  BOD = 10
- (3)  BOD = 5
- (4)  BOD = 20

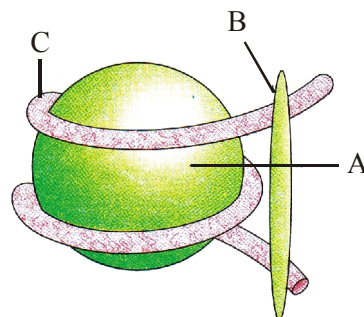
111. Which air borne disease is controlled by use of vaccines ?
- Typhoid
 - Common cold
 - Pneumonia
 - Dengue



Above diagram represent structure of :-

- Cannabinoid molecule
- Morphine
- Cocaine
- Amphetamine

113.



- A-H1 Histone, B-DNA, C-Histone octamer
 - A-Histone octamer, B- H_1 Histone, C-DNA
 - A-DNA, B-Histone octamer, C-H1 Histone
 - A-DNA, B-H1 Histone, C-Histone octamer
114. Which of the following (parts) explants can give rise to a virus-free plant from an infected plant ?
- Leaf
 - Stem
 - Root
 - Apical meristem
115. Which of the following is produced by distillation ?
- Beer
 - Wine
 - Brandy
 - Both (1) and (3)
116. Which type of breeding allows the desirable qualities of two different breeds to be combined ?
- Interspecific hybridisation
 - Cross breeding
 - In-breeding
 - MOET
117. How many attributes of an ecosystem are functional attributes, from the list given below ? Productivity, Species diversity, Decomposition, Nutrient cycling, Dominance, Stratification, Nutrient conservation, Ecological niche, Energy flow
- 3
 - 5
 - 6
 - 7
118. Humming birds are rarely found in polar regions because of :-
- Large surface area
 - Small surface area
 - Larger surface area relative to volume
 - Smaller surface area relative to volume
119. As we move on earth, species diversity increases from :-
- From equator towards pole
 - Mountain base to mountain tops
 - Pole towards equator
 - Both (1) and (2) given above

120. Which of the following has maximum amount of D.D.T. among the options given below ?

- (1) Grass (2) Rabbit (3) Wolf (4) Lion

121. Match the following correctly :-

Column-I		Column-II	
(i)	Tetanus	(a)	Viral disease
(ii)	Shigellosis	(b)	Schick test
(iii)	Rabies	(c)	Risus sardonicus
(iv)	Diphtheria	(d)	Blood & mucus in stool

- (1) i-a, ii-d, iii-b, iv-c (2) i-b, ii-d, iii-a, iv-c
(3) i-d, ii-a, iii-c, iv-b (4) i-c, ii-d, iii-a, iv-b

122. Electrocardiogram is :-

- (1) A machine used to obtain electrical activity of the heart
(2) A graphical representation of the electrical activity of the heart
(3) A machine used to obtain systole and diastole activity of the heart
(4) A graphical representation of systole and diastole of the heart

123. **Statement A** : RNA was the first genetic material.
Statement B : RNA being a catalyst was stable :-

- (1) Both A and B are correct
(2) Only A is correct
(3) Only B is correct
(4) Both A and B are incorrect

124. Choose the incorrect match :-

- (1) Flat bean – Pusa Sem 2
(2) *Brassica* – Pusa Gaurav
(3) SCP – *Spirulina*
(4) Atlas 66 – Transgenic plant

125. The bacteria which help in the breakdown of cellulose and play an important role in nutrition of cattle is _____.

- (1) *Methanobacterium* (2) *Clostridium*
(3) *Aspergillus* (4) *Trichoderma*

126. In absence of honeybees pollination of which of following is affected ?

- (1) Brassica (2) Apple
(3) Sunflower (4) All of these

127. Which of the following is not an example of secondary producer ?

- (1) Phytoplankton (2) Zooplankton
(3) Rabbit (4) Cow

128. CAM pathway is an adaptation in plants to minimise water loss. The type of adaptation is :-

- (1) Morphological (2) Physiological
(3) Behavioural (4) All of the above

129. Which of the following taxa are more vulnerable to extinction ?

- (1) Mammals (2) Insects
(3) Amphibia (4) Birds

130. Which of the following is constituent of Bio Gas?

- (1) Methane (2) Carbon dioxide
(3) Hydrogen sulphide (4) All of the above

131. Interferons are __ (a) __ which are secreted from __ (b) __ :-

- (1) (a)-Glycerides, (b)-Virus infected cells
(2) (a)-Amino acid, (b)-Normal cells
(3) (a)-Proteins, (b)-Normal cells
(4) (a)-Proteins, (b)-Virus infected cells

132. How many of the following traits in garden pea are dominant ?

- Dwarf, Green Seed, Green Pod, Violet flower
(1) 2 (2) 3 (3) 4 (4) 1

133. Which of the following is transcribed by RNA polymerase II in eukaryotes ?

- (1) tRNA (2) rRNA
(3) hnRNA (4) snRNA

134. Which of the following is incorrect for a plasmid ?

- (1) Autonomously replicating
(2) Extra-chromosomal DNA
(3) Circular DNA
(4) Linear DNA

135. Some strains of *Bacillus thuringiensis* produce proteins (toxins) that can kill _____.

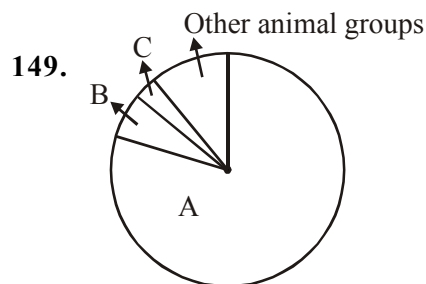
- (1) Lepidopterans (2) Coleopterans
(3) Dipterans (4) All of the above

136. "Hisardale" is a new breed of sheep develop in :-

- (1) Punjab
(2) Bihar
(3) Rajasthan
(4) Delhi

137. Which of the following pair is example of decomposer ?
 (1) BGA and Fungi
 (2) Bacteria and Fungi
 (3) *Cuscuta* and *Gonyaulax*
 (4) Fungi and *Euglena*
138. Which of the following is not an adaptation of altitude sickness ?
 (1) Increase in number of red blood corpuscles
 (2) Decrease in binding capacity of haemoglobin
 (3) Increase in accumulation of lactic acid in muscles
 (4) Increase in breathing rate
139. Extinction of Indigenous cat fishes in our rivers due to introduction of African catfish, is example of extinction caused due to :-
 (1) Habitat loss
 (2) Over exploitation
 (3) Alien species invasions
 (4) Co-extinctions
140. Which of the following disaster is not related to radioactive wastes ?
 (1) Three Mile Island disaster
 (2) Chernobyl disaster
 (3) Bhopal Gas Tragedy
 (4) Both (1) and (2) given above
141. Which of the following is the example of passive immunisation ?
 (1) Toxoids (2) RDT vaccines
 (3) Anti toxins (4) Venom
142. How many types of gametes will be produced by a plant having genotype $AaBbDdEE$?
 (1) 16 (2) 8 (3) 4 (4) 2
143. In lac operon lac y gene codes for ____ :-
 (1) β -galactosidase (2) Permease
 (3) Transacetylase (4) Hexokinase
144. ECORI always cut DNA molecules at a particular point by recognising a specific sequence of ____ base pairs.
 (1) 6 (2) 3 (3) 4 (4) 8
145. Silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of mRNA is known as _____.
 (1) Transformation (2) Translation
 (3) RNA interference (4) RNA induction
146. Which of the following disease caused by protozoa in cattle ?
 (1) Ranikhet (2) Bird flu
 (3) Babesiosis (4) Anthrax

147. In an ecosystem as interlinking between food chains increases, stability of the ecosystem :-
 (1) Increases
 (2) Decreases
 (3) First increases then decreases
 (4) First decreases then increases
148. If detritus is rich in lignin and chitin, rate of decomposition in a particular set of environmental condition, will be :-
 (1) Faster
 (2) Slower
 (3) Constant
 (4) Firstly faster then become slower after some time



Above pie chart represents proportionate number of species of invertebrates, A, B and C represents:-

- (1) A = Insects, B = Crustaceans, C = Molluscs
 (2) A = Insects, B = Molluscs, C = Crustaceans
 (3) A = Molluscs, B = Crustaceans, C = Insects
 (4) A = Molluscs, B = Insects, C = Crustaceans
150. Which of the following statement(s) is/are correct among the options give below ?
 (A) Domestic sewage primarily contains biodegradable organic matter.
 (B) High concentration of DDT promotes calcium metabolism in birds.
 (C) Eutrophication is the natural ageing of lake.
 (D) Integrated waste water treatment includes mix of artificial and natural processes.
 (1) A, B, C and D (2) A, C and D
 (3) B, C and D (4) C and D
151. Physical barrier and MALT are present in :-
 (1) Lungs only
 (2) GI tract and urogenital tract only
 (3) Respiratory tract and urogenital tract only
 (4) Respiratory tract, urogenital tract and GI tract

152. What is the F_2 phenotypic ratio in a dihybrid cross involving the genes showing dominant epistasis?
 (1) 15 : 1 (2) 9 : 7
 (3) 12 : 3 : 1 (4) 9 : 3 : 4
153. The last step in DNA finger printing technique is :-
 (1) Autoradiography
 (2) Digestion of DNA by restriction endonuclease
 (3) Separation of DNA fragments by electrophoresis
 (4) Hybridisation using labelled VNTR probe
154. Ethidium bromide stained agarose gel is exposed to _____, to observe DNA as bright orange colored bands.
 (1) Air (2) UV light
 (3) Infra-red light (4) Green light
155. Which of the following peptide is present in mature human insulin ?
 (1) A-peptide (2) B-peptide
 (3) C-peptide (4) Both (1) and (2)
156. The minimum level in ecological hierarchy in which abiotic factors are described along with biotic factors :-
 (1) Population (2) Organism
 (3) Community (4) Ecosystem
157. Which of the following statement is not correct about Detritus Food Chain (DFC) ?
 (1) In DFC energy flow is slower than Grazing Food Chain (GFC).
 (2) Bacteria and fungi are primary consumers in DFC.
 (3) DFC do not follow 10 percent energy limitation.
 (4) DFC is based on predation only.
158. Which of the following is not an example of sedimentary cycle ?
 (1) Carbon cycle (2) Phosphorus cycle
 (3) Calcium cycle (4) Sulphur cycle
159. Which of the following gaseous pollutant causes destruction of chlorophyll ?
 (1) CFC (2) Carbon mono oxide
 (3) Sulphur di oxide (4) Ethylene
160. Which of the following option represents correct matching pairs ?
 (A) National Environment Act (i) 1981
 (B) Air Act (ii) 1986
 (C) Montreal protocol (iii) 1997
 (D) Kyoto protocol (iv) 1987
 (1) A-i, B-ii, C-iii, D-iv (2) A-ii, B-i, C-iii, D-iv
 (3) A-ii, B-i, C-iv, D-iii (4) A-iii, B-iv, C-i, D-ii
161. How many statements are correct regarding Cancer?
 (a) Taxol is obtained from *Taxus baccata*
 (b) Biopsy is a histopathological study
 (c) X-rays causes neoplastic transformation
 (d) Alkaline phosphatase is a tumor marker for prostate cancer
 (1) Two (2) Three (3) One (4) Four
162. If a red flowered snapdragon plant is crossed with a pink flowered plant, then what will be the phenotypic ratio in the progeny ?
 (1) 2 : 1 (2) 3 : 1
 (3) 9 : 3 : 3 : 1 (4) 1 : 1
163. In humans how many genes are present on chromosome number 1 ?
 (1) 2698 (2) 9968 (3) 2968 (4) 2896
164. Which of the following enzymes is not used for isolation of DNA from bacteria, plant or fungi ?
 (1) Lysozyme (2) Chitinase
 (3) Cellulase (4) DNase
165. Which human protein is used to treat emphysema ?
 (1) α -1-antitrypsin (2) α -1-lactalbumin
 (3) α -amylase (4) β -galactosidase
166. Which of the following characteristic of succession does not show an increase from pioneer towards climax state :-
 (1) Niche specialisation
 (2) Humus content
 (3) Net community productivity
 (4) Biomass
167. In an ecosystem, pyramid of number represents:-
 (1) Standing state
 (2) Biotic potential
 (3) Productivity
 (4) Both (1) and (2) given above
168. Which of the following is a reason of higher biodiversity in tropical regions ?
 (1) More seasonal variation
 (2) Lesser solar availability
 (3) Lesser rainfall
 (4) Long evolutionary time
169. Device which is based on precipitation of dust particles on the basis of electrostatic force of attraction :-
 (1) Dry scrubber
 (2) Electrostatic precipitator
 (3) Wet scrubber
 (4) Cyclonic separators

170. If a habitat is just being established then, most important factors determining population size :-
 (1) Emigration (2) Immigration
 (3) Birth rate (4) Death rate

171. Match the following regarding STDs :-

STD		Pathogen	
(i)	Gonorrhoea	(a)	Protozoan
(ii)	Genital warts	(b)	Virus
(iii)	Trichomoniasis	(c)	Bacterium
(iv)	Syphilis	(d)	T.pallidum

- (1) (i)-d, (ii)-a, (iii)-b, (iv)-c
 (2) (i)-c, (ii)-b, (iii)-a, (iv)-d
 (3) (i)-a, (ii)-b, (iii)-c, (iv)-d
 (4) (i)-b, (ii)-c, (iii)-d, (iv)-a

172. Who noted that the behaviour of chromosomes was parallel to the behaviour of genes ?

- (1) Sutton (2) Correns
 (3) Boveri (4) Both (1) and (3)

173. Which of the following is the root of any breeding programme ?

- (1) Collection of Variability
 (2) Evaluation and selection of parents
 (3) Selection and testing of superior recombinants
 (4) Cross hybridisation among the selected parents

174. How many DNA molecules will be obtained from one DNA molecule after 3PCR (Polymerase chain reaction) cycles ?

- (1) 8 (2) 16 (3) 6 (4) 32

175. The use of bioresources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment is known as :-

- (1) Bio-war (2) Biopiracy
 (3) Bio-cloning (4) Bio-processing

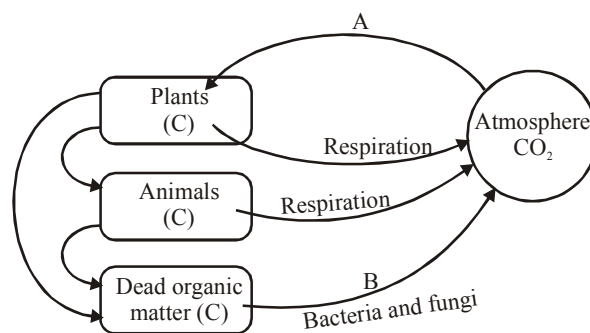
176. Which of the following is not the special adaptation evolved in parasites ?

- (1) Presence of well developed digestive system
 (2) Loss of unnecessary sense organs
 (3) Presence of adhesive organs
 (4) High reproductive capacity

177. In a grassland ecosystem, which is the ecologically strongest among the options given below :-

- (1) Lion (2) Wolf (3) Rabbit (4) Jackal

178.



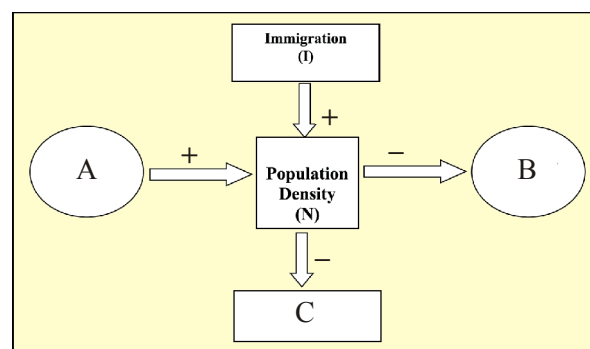
In the above diagram, process A and B are respectively :-

- (1) A = Photosynthesis, B = Chemosynthesis
 (2) A = Respiration, B = Photosynthesis
 (3) A = Photosynthesis, B = Oxidation
 (4) A = Decomposition, B = Photosynthesis

179. Which of the following ozone depleting substance also causes green house effect ?

- (1) CO₂ (2) CFC
 (3) CH₄ (4) All of the above

180.



In the digram given above, which of the following option correctly represents A, B and C.

- (1) A = Death rate, B = Birth rate, C = Emigration
 (2) A = Birth rate, B = Death rate, C = Emigration
 (3) A = Emigration, B = Death rate, C = Birth rate
 (4) A = Emigration, B = Death rate, C = Birth rate

SPACE FOR ROUGH WORK

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with Paper code and Your Form No.

Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with Paper code and Your Form No.

Corporate Office : ALLEN CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

+91-744-2757575 **info@allen.ac.in** **www.allen.ac.in**